AI 应用落地方案说明书

1. Project background and objectives
    1.1. DCC：Document Control Center
    1.2. DCN：Document Change Notice(for Company Rules and Regulations)
    1.3. DMS：Document Management System
    1.4. Background description

    With the continuous evolution of artificial intelligence technology, large language models (LLM) and multimodal models have demonstrated remarkable capabilities in comprehension, generation, and reasoning. An increasing number of enterprises are beginning to experiment with integrating AI technology into their actual business processes, aiming to enhance information processing efficiency, optimize decision-making quality, and reduce reliance on human experience. Against this backdrop, the focus of AI application has gradually shifted from "model effect demonstration" to "systematic implementation capability". How to embed model capabilities into existing business systems in a stable and controllable manner has become a common concern for both technical and business teams.

    1.4.1. Industry prospects1
    1.4.2. Industry prospects2
    1.4.3. Industry prospects3
    1.4.4. Industry prospects4
    1.4.5. Industry prospects

    The current industry as a whole exhibits the following characteristics：

    - The capabilities of pre-trained models have rapidly improved, but there are still differences in scenario generalization
    - The scale of internal data within the enterprise continues to grow, leading to a strong demand for intelligent retrieval and analysis
    - AI systems are gradually evolving from single-point applications towards platformization and servitization

    1.4.6. Main pain points

    In the actual implementation process, common issues include but are not limited to：

    - The data sources are complex, with a coexistence of structured, semi-structured, and unstructured data
    - The coupling between model services and business logic is too high, resulting in poor system maintainability
    - Without a unified evaluation and monitoring mechanism, it is difficult to continuously ensure the effectiveness of the model

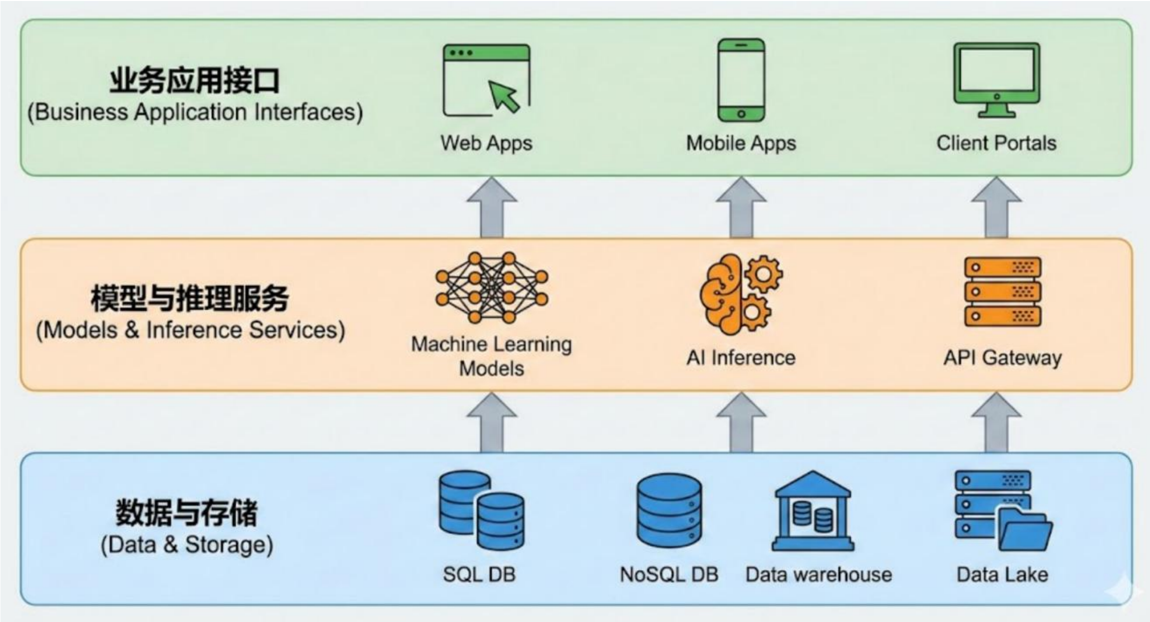
    1.5. Construction objectives

    A. The goal of this project is to build an AI application system that is scalable, maintainable, and implementable,
    B. Capable of supporting multiple business scenarios within the enterprise,
    C. Provide a technical foundation for subsequent iterations。

2. Overall scheme design

2.1. Overview of system architecture

The system is designed with a layered architecture, encompassing a data layer, a model layer, and an application layer. This design ensures the independence between modules and facilitates the addition of new functional modules in the future.



2.2. 技术选型说明

2.2.1. 后端与基础设施

- 编程语言：Python
- 服务框架：FastAPI
- 模型部署：vLLM / Triton
- 容器与编排：Docker + Kubernetes

2.2.2. 模型相关

- 大语言模型（LLM）
- 向量模型（Embedding）
- RAG 检索增强生成

2.2.3. 其他模型相关

- 视觉大模型（VLM）
- 向量模型（Embedding）
- RAG 检索增强生成

2.3. Technical selection description

2.3.1. Backend and Infrastructure

- Programming language: Python
- Service framework: FastAPI
- Model deployment: vLLM / Triton
- Containers and orchestration: Docker + Kubernetes

2.3.2. Model-related

- Large Language Model (LLM)
- Vector model (Embedding)
- RAG: Retrieval Augmented Generation

2.3.3. Other model-related

- Visual Large Model (VLM)
- Vector model (Embedding)
- RAG: Retrieval Augmented Generation

2.3.4. Backend and Infrastructure

- Programming language: Python
- Service framework: FastAPI
- Model deployment: vLLM / Triton
- Containers and orchestration: Docker + Kubernetes

2.3.5. Model-related

- Large Language Model (LLM)
- Vector model (Embedding)
- RAG: Retrieval Augmented Generation

2.3.6. Other model-related

- Visual Large Model (VLM)
- Vector model (Embedding)

2.3.7. **RAG: Retrieval Augmented Generation**Other model-related

- Visual Large Model (VLM)
- Vector model (Embedding)
- RAG: Retrieval Augmented Generation

2.3.8. Other model-related

- Visual Large Model (VLM)
- Vector model (Embedding)

- RAG: Retrieval Augmented Generation
- Service framework: FastAPI

3. Core functional modules

3.1. All documents generated by function departments should be named according to the below-mentioned categories of documents.

3.1.1. Data Source

A. Equipment pperation instruction

Data Type	**Data Type**	**Data Type**	**Data Type**
Document Data	Document Management System	PDF / DOCX	Daily
Tabular Data	Business Databases	CSV / XLSX	Real-time
Log Data	Application Systems	JSON	Real-time

B. Other documents

Data Type	**Data Type**	**Data Type**	**Data Type**
Document Data	Document Management System	PDF / DOCX	Daily
Tabular Data	Business Databases	CSV / XLSX	Real-time
Log Data	Application Systems	JSON	Real-time

3.1.2. All documents generated by function de

3.1.3. 数据处理模块包括以下流程

- 清洗无效数据，去除空值与重复数据
- 结构化与切分（Chunking）
- 向量化与索引构建，用于快速检索和匹配

3.2. 模型服务模块

3.2.1. 推理服务

模型通过统一的 API 服务对外提供能力，支持高并发和流式输出，以满足业务实时性要求。

3.2.2. 模型更新策略

- 支持模型热更新，避免服务中断
- 支持多版本并行部署，便于 A/B 测试和回滚

3.3. 应用层模块

3.3.1. 典型应用场景

- 智能问答助手

- 文档检索与总结
- 业务辅助决策

3.3.2. 权限与安全

- 用户身份鉴权
- 请求审计与日志记录
- 数据访问控制

4. 4. Implementation plan and milestones

4.1. Categories and hierarchy of documents

Task ID	Task Des	Priority	Assigned Department	Estimated Hours	Status
T-001	Initial Client Consultation	High	Sales	4	Completed
T-002	Requirement Analysis	Critical	Engineering	12	In Progress
T-003	Market Competitor Audit	Medium	Marketing	8	Pending

4.2. 实施阶段划分

Task ID	Task Des	Priority	Assigned Department	Estimated Hours	Status
T-001	Initial Client Consultation	High	Sales	4	Completed
T-002	Requirement Analysis	Critical	Engineering	12	In Progress
T-003	Market Competitor Audit	Medium	Marketing	8	Pending

4.3. 风险与应对

4.3.1. 技术风险

- 模型效果不稳定 → 引入评估与回滚机制

- 系统性能瓶颈 → 采用分布式部署与负载均衡

4.3.2. 业务风险

- 需求变更频繁 → 采用模块化设计与灵活迭代
- 数据安全问题 → 强化访问控制和日志审计

5. Attachment
   5.1. Atachment1：Formate for xxxxx1
   5.2. Atachment1：Formate for xxxxx2
   5.3. Atachment1：Formate for xxxxx3
   5.4. Atachment1：Formate for xxxxx4
   5.5. Atachment1：Formate for xxxxx5

6. 总结与展望
   6.1. Effective period of TECN

SMIC document security levels are categorized accounting to the security control level's defined in "CR-GWNL-01-1001 SMIC Classified Information Protection Policy (CIPP)"as follows

Security B-SMIC Confidential

Security C-SMIC Restricted

6.2. Effective period of TECN
   A. Document format should follow the requirements sign-off
   B. Document sign-off should follow the requirements in item 20
   C. Document access authority should follow the requirements in item 18.2
   D. Document effectiveness cycle time should be within 14 working days

6.3. All documents generated by function departments should be named according to the below-mentioned categories of documents for example,"SMIC Document Control Procedure" Document title naming rules are follows:

6.3.1. FAB documents

A. Equipment operation instruction

Document Scope	Naming Rule Format
200mm Fab	SMIC 200mm XXX(Module)XXX(Equipment/System Name)O.I.E.g SMIC 200mm ETCH LAM TCP 9400 DFMP.M. O.I.
300mm Fab	SMIC 300mm XXX(Module)XXX(Equipment/System Name)O.I.E.g SMIC 300mm Etch AMAT eMAX O.I.

ALL Fab :Moduleis operational	SMIC XXX(Module)XXX(Equipment/System Name)O.I.E.g SMIC LITHO Entegris XCDA PM O.I.

B. Other documents

Document Scope	Naming Rule Format
200mm Fab	SMIC 200mm XXX(Module)XXX(Function/System Name)Rule/Procedure E.g. SMIC 200mm MFG Production Plan Review Procedure
300mm Fab	SMIC 300mm XXX(Module)XXX(Function/System Name)Rule/Procedure E.g. SMIC 300mm YE Recipe Request Procedure
ALL Fab :Moduleis operational	SMIC XXX(Module)XXX(Function/System Name)Rule/Procedure E.g. SMIC PIE PCM(Process Control Monitor) Specification Control Procedure

6.3.2. Non-FAB documents/DCN
(E.g. Mask Operation,RE lab,Facility,etc)

Document Scope	Naming Rule Format
SMIC(SH,BJ,TJ,SZ)	SMIC (XX)(Site) XXX(Function area)XXX(Equipment/System name) O.I./Rule/Procrdure/Guideline/Policy E.g. SMIC (SH/BJ) Information Equipment Management Protection Procedure
SMIC and SMIC managed facilities	SMIC XXX(Function Area)XXX(Function/Equipment/System Name) O.I./Rule/Procedure E.g. SMIC Facility UPS O.I.

6.3.3. Technology development related documents naming rule is defined in " TD-GENL-01-2047 SMIC Deaign Rules Template" and "TD-GENL-01-2064 SMIC PDK Document Title and Attached File Naming Rule"